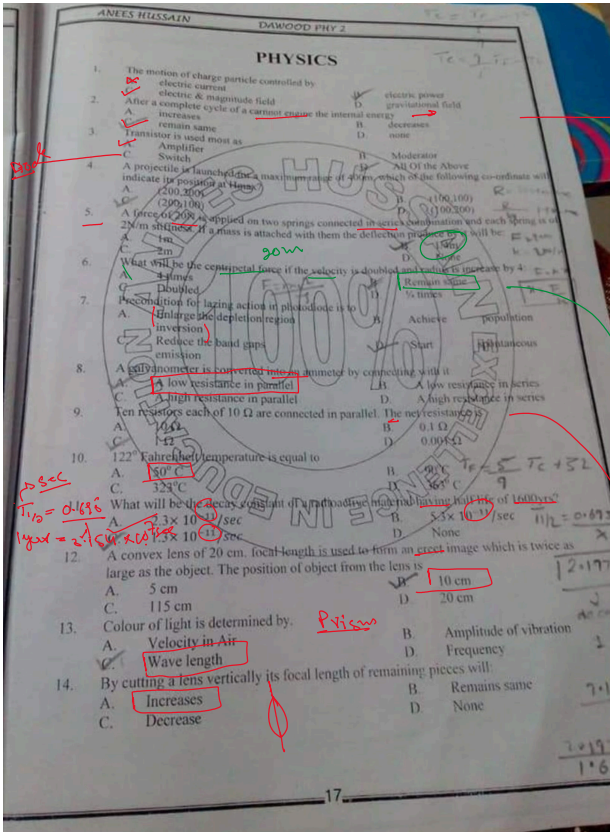


DUET PAST PAPERS-2

Friday, 26 July 2024 9:31 am



electron proton.

ideal engine efficiency 100%

$K_1 = 20 \text{ N/m}$
 $K_2 = 20 \text{ N/m}$
 $F = 20 \text{ N}$
 $\frac{1}{K} = \frac{1}{K_1} + \frac{1}{K_2} = \frac{1}{20} + \frac{1}{20} = \frac{2}{20} = \frac{1}{10}$
 $K = 10 \text{ N/m}$
 $x = \frac{F}{K} = \frac{20}{10} = 2 \text{ m}$
 $F = \frac{mv^2}{r} = \frac{m(2v)^2}{4r} = \frac{4mv^2}{4r} = \frac{mv^2}{r}$

Galvanometer $\rightarrow$ ammeter

$R_1 \text{ --- } R_2 = 10 \Omega$ (Parallel)

$\frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_n} = \frac{1+1+\dots+1}{10} = \frac{10}{10} = 1$
 equal resistors.

Parallel $\frac{1}{R_n} = \frac{10}{10} = 1 \Omega$
 Series $n \times R_1$

$102 \Omega \mid C = \frac{5}{9}(F-32) \mid C = \frac{5}{9}(80) \mid \frac{9C}{5} = 160$
 $C = \frac{5}{9}(102-32) \mid C = 50 \Omega \mid \frac{9C}{5} = 90$

$\lambda = ?$
 $T = 1600 \text{ year} = 5.046 \times 10^8 \text{ sec}$

$T_2 = 0.695 \Rightarrow \lambda = \frac{0.695}{T_2} = \frac{0.695}{5.046 \times 10^8} = 1.376 \times 10^{-9} \text{ m}$

$f = 20 \text{ cm}$
 $M = \frac{q}{P}$
 $2 = \frac{q}{P}$
 $2P = 20$
 $P = 10 \text{ cm}$



$20 = \frac{q}{P}$
 $10 - (-5) = 15 \text{ m/s}$
 $u = ?$
 $u = 150$
 $v = \frac{u}{\lambda} \mid \lambda = \frac{150}{10} = 15$
 $15 = \frac{150}{\lambda} \mid \lambda = 10 \text{ sec}$

